UNIVERSITY OF GOTHENBURG

INSTITUTIONEN FÖR FILOSOFI,
LINGVISTIK OCH VETENSKAPSTEORI

Beyond Pixels and Words : Frames of Reference and Priming in Vision Language Models

Tianyi Gong

Essay/Thesis:	Masteruppsats, 30 hp
Programme/Course:	Språkteknologi, masterprogram, 120 högskolepoäng
Level:	Avancerad nivå
Semester:	Spring, 2026
Supervisor:	Simon Dobnik
Examiner:	
Keywords:	Vision Language Model,Visual Reasoning, Spatial Reasoning

Abstract

We investigate the ability of vision language models (VLMs) to reason about spatial relations from different Frames of Reference (FoRs) and whether their FoR preferences can be influenced through priming. We evaluate three VLMs using manually constructed scenes in SketchUp. The results show that all three models strongly prefer the speaker's perspective. Some models can shift between speaker and hearer perspectives, but they struggle to adopt the intrinsic FoR. Meanwhile, although priming can occasionally induce the intrinsic FoR, this effect is not maintained in the subsequent turns, with the models tending to return to the speaker's perspective. These findings suggest that VLMs' spatial reasoning is strongly anchored to the initial perspective and that intrinsic FoR remains particularly challenging for current VLMs.

Acknowledgements

I am grateful for the time I spent in Sweden. It was a peaceful and profoundly meaningful period of my life.

Contents

1 Introduction

Expressions such as *right of*, *left of*, and *in front of* are commonly used to describe the spatial location of the objects in our daily life. However, the spatial relationships are not fixed but determined by several factors. Based on Levinson, 2003, spatial relations can be established based on different coordinate systems, including the observer's viewpoint (relative FoR), the intrinsic orientation of the reference object (intrinsic FoR), or fixed environmental directions (absolute FoR).

As vision language models (VLMs) have rapidly advanced in recent years, they have shown impressive capabilities in spatial reasoning. However, most existing VLMs are primarily trained on datasets that spatial information is often observed from a single viewpoint. For instance, the classic CLEVR dataset (Johnson et al., 2017) generates 3D scenes from a single fixed camera viewpoint, where spatial relations are described based on the rendered visual perspective. Such dataset limitations may restrict models' ability to perform spatial reasoning under different frames of reference.

There are several recent studies related to spatial reasoning in different FoR. Liu et al. (Liu, Emerson, and Collier, 2023) introduce the Visual Spatial Reasoning (VSR) dataset, which evaluates whether VLMs can understand spatial relations in images. The dataset contains spatial expressions involving different reference perspectives, including intrinsic and viewpoint interpretations. Similarly, Lee et al. (Lee et al., 2022) create a dataset of visual questions involving spatial relations, where some questions using an intrinsic FoR and others using the camera viewpoint. However, these benchmarks mainly evaluate spatial reasoning under predefined settings. The ability of VLMs to flexibly reason across different frames of reference remains to be explored.

A Human study further shows that spatial perspectives are flexible rather than fixed. Dobnik et al. (Dobnik, Koniaris, and Kelleher, 2013) investigate the role of priming and perspective alignment in dialogue, showing that speakers may adapt their choice of spatial reference frames based on context.

Inspired by these findings, we decide to investigate the ability of VLMs to reason about spatial relations in different kinds of frame of reference (FoR) and further investigate whether their FoR preference can be influenced through priming.

To achieve our goals, we generate a small set of 3D scenes with strictly controlled object positions by SketchUp for our experiments. Each scene contains objects with clear intrinsic orientations, such as sofas, chairs, tables and lamps.

2 Experiemnt

2.1 Experiment Design

We design two tasks here to answer two different question: Can VLLM understand difference between two kinds of FoR?(relative and intrinsic) and Does dialogue priming influence the FoR?

Therefore, the first task examines whether MLLMs can distinguish different FoR. We keep the scene constant and manipulate only the prompt. The model is asked the same spatial question three times about the same image from three different perspective, each time under a different instruction. For instance, as shown in 2.1, in the first condition, the question is grounded in the speaker perspective (e.g. "From your perspective looking at the scene, what is to the left of the sofa?"). In the second condition, it is grounded in the intrinsic FoR of an object (e.g. "From the sofa's own perspective, what is to the left of the sofa?"). In the third condition, the model is asked to adopt a hearer's perspective by being told that the interlocutor is standing opposite, facing the scene from 180° (e.g. "I am standing opposite you, facing you. From my perspective, what is to the left of the sofa?").

The second task is designed to solve the second question. In this task, each model is presented with a scene and a 4-turn vision question answering sequence. In Turn 1, a spatial description grounded in the intrinsic FoR is provided as context (e.g. There is a TV in front of the sofa), implicitly establishing the object's intrinsic orientation as the reference frame. In Turns 2 and 3, the model is asked an ambiguous spatial question whose answer differs depending on which FoR is applied. (e.g. Is there a chair to the left of the sofa? In this case, it's the spearker's or intrinsic perspective). It's worth noting that both chairs and both tables satisfy the condition, but under different FoRs. To distinguish which object the model is referring to, the two objects of the same type are assigned different colors. The model is additionally required to specify the color of the object it refers to. In Turn 4, the model is asked to freely describe the location of an object : Where is the wardrobe? Please describe its location. The generated description is analysed to determine which FoR the model adopts, and whether this choice reflects the frame established in Turn 1 or defaults to an alternative frame such as the relative FoR.

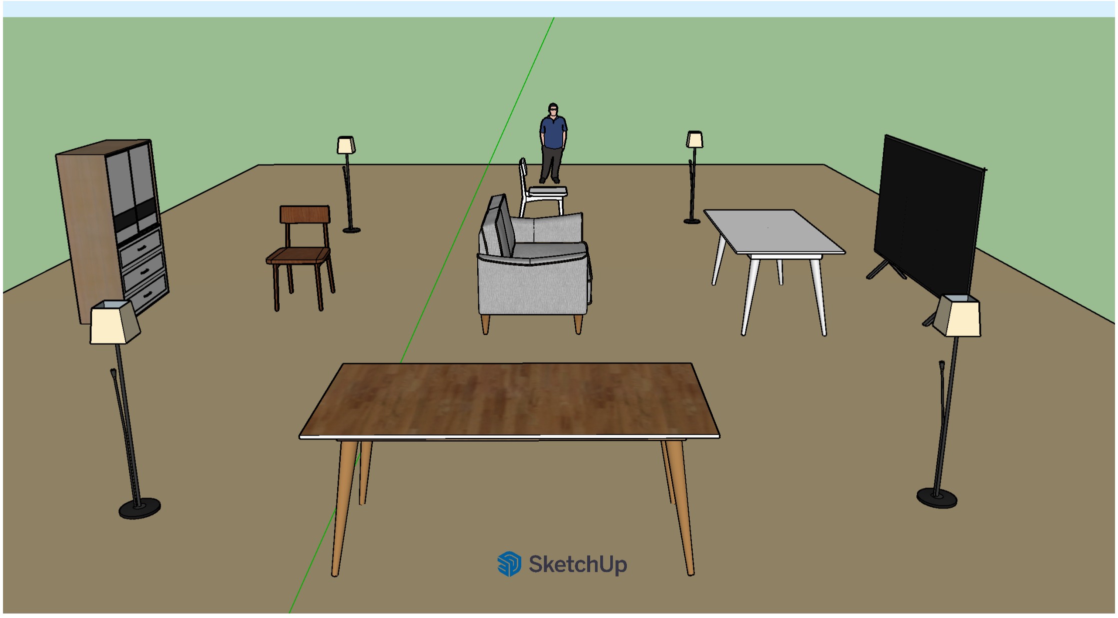

Figure 2.1: Example of a 3D scene generated in SketchUp.

2.2 Dataset

There are several constraints on the scenes that should be noted. First, the object located at the center of the scene should have a clear intrinsic orientation, allowing spatial relations to be interpreted under an intrinsic FoR. Meanwhile, there are two special objects in each scene (this is not necessarily the case, as the blue shirt person on the opposite side – the hearer from the model's perspective, can also be used for priming). As shown in Figure 2.1, they are TV and wardrobe. One object is used for priming the model with a specific frame of reference, the other is used to evaluate which frame of reference the model prefers after a 4-turn dialogue.

Furthermore, the positions of objects are strictly controlled to create situations where different FoR may lead to different interpretations. For example, one chair can be placed to the left of a sofa from the speaker's perspective, while another chair can be placed to the left of the sofa according to the sofa's intrinsic orientation. As we said above, to distinguish between these same category objects and determine which spatial relation the model refers to, objects with identical categories but different spatial roles are assigned different colors.

2.3 Models

Three models are tested in our experiments: Qwen2.5-VL-3B, Gemini-3.1-Flash-Lite, and Kimi 2.6. Qwen2.5-VL-3B Bai et al., 2025 is an open-source vision language model with 3 billion parameters. In our experiments, it is deployed locally through Ollama.

Gemini-3.1-Flash-LiteGoogle, 2026 and Kimi 2.6Moonshot AI, 2026 are accessed through APIs.

3 Results

3.1 Task1

	Speaker	Intrinsic	Hearer
Qwen	T\T\F	F(Speaker)\F(Speaker)\F(Speaker)	F(Speaker)\F(Speaker)\F(Speaker)
Gemini	T\T\T	F(Speaker)\F(Speaker)\F(Speaker)	T\T\T
Kimi	T\T\T	F(Speaker)\F(Speaker)\F(Speaker)	F(Speaker)\T\T
T denotes a correct (true) response, while F denotes an incorrect (false) response.			

Table 3.1: Results of Qwen, Gemini, and Kimi under different frames of reference.

Table 3.1 shows the results of Task 1. As shown in the table, Qwen only succeeds in answering questions from the speaker's perspective, while it fails on questions requiring either an intrinsic or a hearer's perspective, suggesting that Qwen struggles with perspective shifts and tends to rely on the speaker's perspective when answering spatial questions.

Gemini performs obviously better than Qwen, which successfully answer questions from both the speaker's and the hearer's perspectives. However, it still fails when the perspective shifts to the object's own perspective – intrinsic FoR.

Kimi shows a performance similar to Gemini and it produces one additional false response under the hearer's perspective.

Almost all of the false responses are answered according to the speaker's perspective, suggesting that the models tend to default to the speaker's perspective when they fail to adopt the required perspective.

3.2 Task2

Table 3.2 shows the results of Task 2. We surprisingly find that all three models consistently rely on the speaker's perspective. During Turn 1, which serves as the priming turn, Qwen is able to correctly identify the intrinsic Frame of Reference (FoR). In contrast, both Gemini and Kimi struggle to recognize the intrinsic orientation of the object, with Gemini correctly identifying it only once.

For instance, Gemini succeeded in the first right-priming trial in Scene 1, stating: "Yes, that is correct. Based on the image, the television is positioned in front of the gray

	T1	T2	T3	T4
Qwen	I (success)\I (success)\I (success)	S\S\S	S\S\S	S\S\S
Gemini	I (success)\I (fail)\I (fail)	S\S\S	S\S\S	S\S\S
Kimi	I (fail)\I (fail)\I (fail)	S\S\S	S\S\S	S\S\S
I denotes intrinsic, while S denotes speaker.				

Table 3.2: Perspective preferences of the models across different tasks.

sofa, situated towards the right side of the room." However, it failed in Scene 2, stating: "I can see a person in a blue shirt in the image, but they are positioned behind the bed, not to the right of it." It seems like that sofa has a more salient and clear interpretable directionality than beds, which can be more easily accepted by the model.

4 Conclusion

In this paper, we investigated the ability of VLMs to reason about spatial relations under FoR, with a particular focus on whether their perspective preference could be influenced through priming.

We evaluated three models, Qwen, Gemini, and Kimi across two tasks including questions from speaker's, hearer's, and intrinsic perspectives.

The results show that the models exhibit a strong preference for the speaker's perspective. Task 2 further suggests that the models exhibit a strong resistance to adopting the intrinsic FoR, even when it is introduced through priming.

Overall, our findings suggest that current VLMs are capable of reasoning about some spatial relations, but their performance is strongly based on the perspective we choose. In particular, intrinsic FoR appears to be obviously more difficult to recognize. Although there are a few successful cases in which priming leads the models to adopt the intrinsic FoR, this shift is not maintained in subsequent turns, where the models return to the speaker's perspective immediately . This suggests that their spatial reasoning is strongly anchored to the initial perspective. It seems like that models have a tendency to fall back on the speaker's perspective when a different FoR is required. Some models are capable of shifting between the speaker's and hearer's perspectives (Gemini), but they show a much greater difficulty in adopting the intrinsic FoR.

5 Future work

A limitation of this study is the small scale of the experiments because the test scenes were manually constructed, therefore the number of evaluated scenes was limited. Future work could develop an automated procedure for generating and evaluating spatial scenes, allowing a larger and more diverse set of scenes to be tested. We believe it would provide more comprehensive data and make it possible to examine the models' FoR preferences across a wider range of spatial configurations.

Bibliography

Bai, Shuai et al. (2025). "Qwen2.5-VL Technical Report". In: *arXiv preprint arXiv:2502.13923*.

Dobnik, Simon, Christos Koniaris, and John D. Kelleher (2013). "Investigating the Role of Priming and Alignment of Perspective in Dialogue". In: *Proceedings of the 17th Workshop on the Semantics and Pragmatics of Dialogue*. Amsterdam, The Netherlands.

Google (2026). *Gemini API Documentation*. URL: `https://ai.google.dev/gemini-api/docs`.

Johnson, Justin, Bharath Hariharan, Laurens van der Maaten, Li Fei-Fei, C. Lawrence Zitnick, and Ross Girshick (2017). "CLEVR: A Diagnostic Dataset for Compositional Language and Elementary Visual Reasoning". In: *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*, pp. 2901–2910.

Lee, Jae Hee, Matthias Kerzel, Kyra Ahrens, Cornelius Weber, and Stefan Wermter (2022). "What is Right for Me is Not Yet Right for You: A Dataset for Grounding Relative Directions via Multi-Task Learning". In: *Proceedings of the Thirty-First International Joint Conference on Artificial Intelligence*, pp. 1039–1045.

Levinson, Stephen C. (2003). *Space in Language and Cognition: Explorations in Cognitive Diversity*. Cambridge: Cambridge University Press.

Liu, Fangyu, Guy Emerson, and Nigel Collier (2023). "Visual Spatial Reasoning". In: *Transactions of the Association for Computational Linguistics* 11, pp. 635–651.

Moonshot AI (2026). *Kimi API Documentation*. URL: `https://platform.kimi.ai/docs/models`.